

LAB 01

Cloud Computing

LAB 1 SOLUTION

Name: Noor Ul Ain

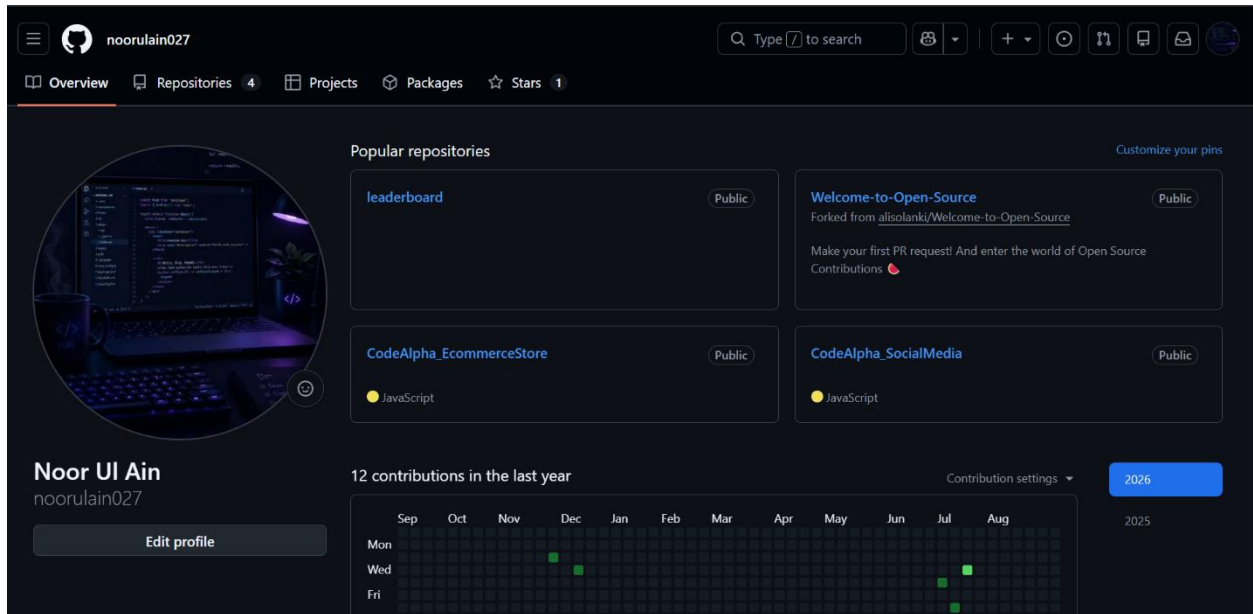
Registration No.: FA24B1-SE-027

Course Code: kjsrxxoo

Section: A

GitHub Username: noorulain027

Task no 1: Create and Configure Your GitHub Account

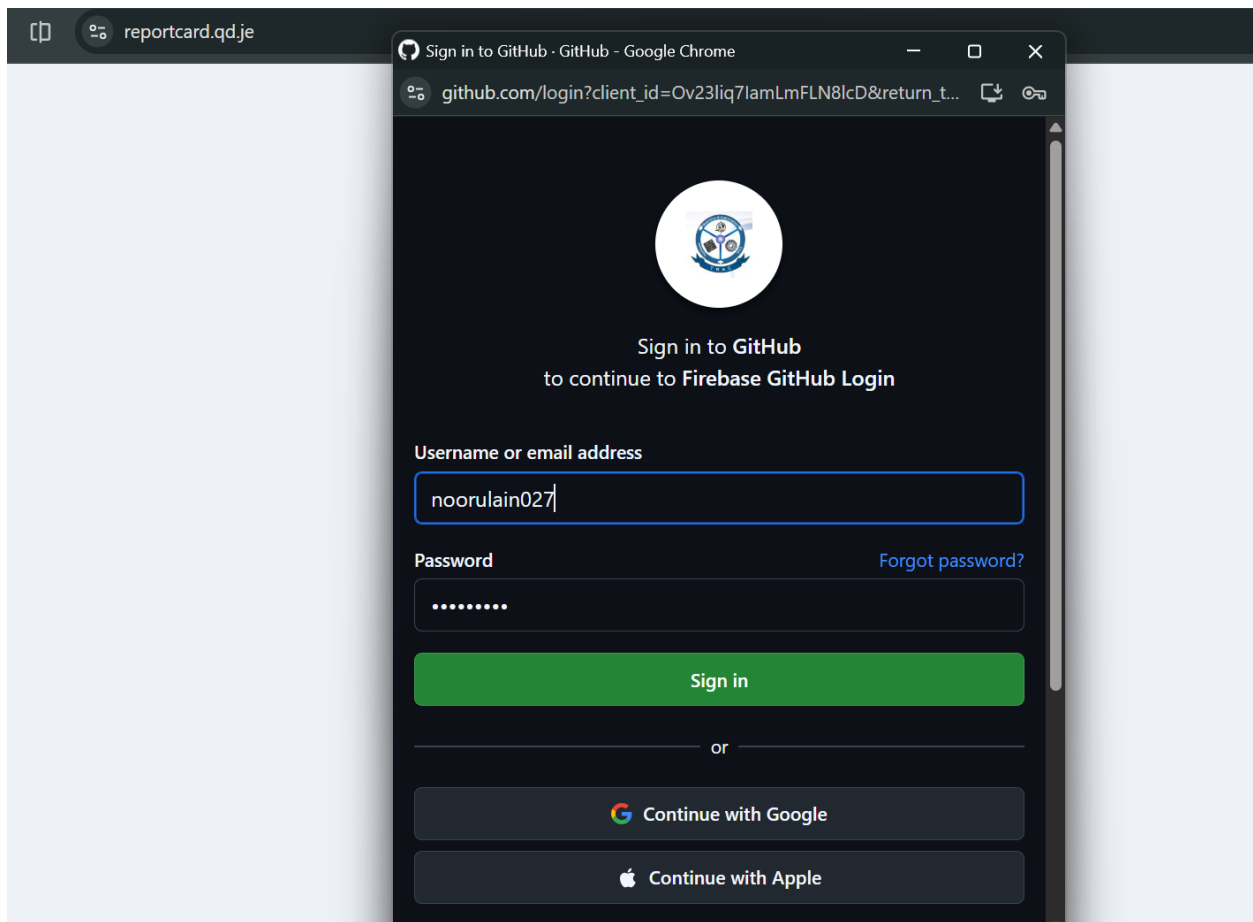


Shows GitHub profile

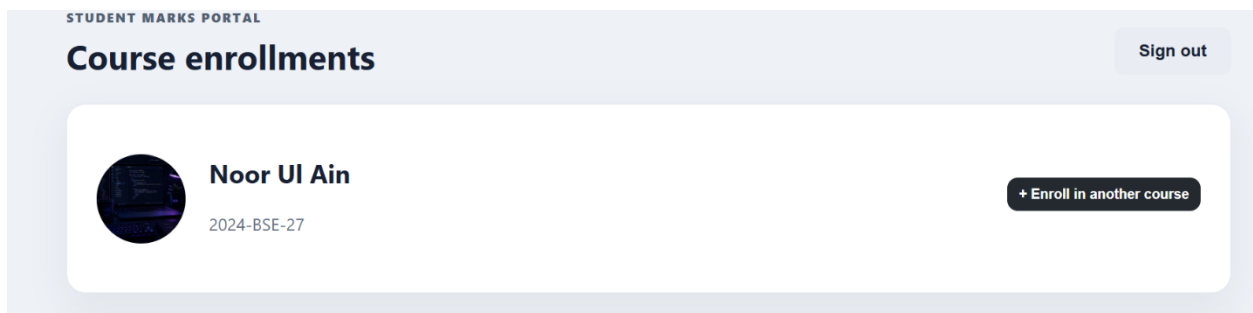


Shows GitHub Actual Name

Task 2: Register in the Student Marks Portal



Shows Student Portal Login Page



Shows Portal GitHub Profile

ENROLLMENT STATUS

Your courses

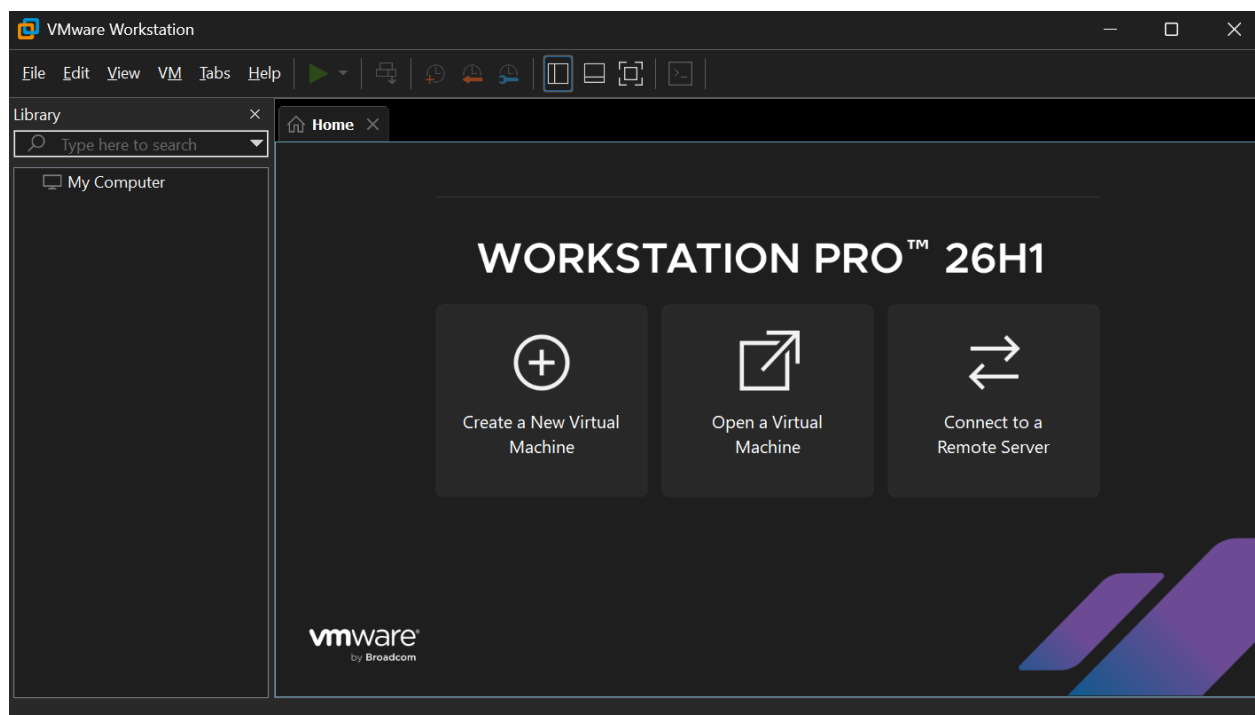
Cloud Computing (SE-665) Section A

Waiting for administrator approval

Pending approval

Shows Portal enrollment Status

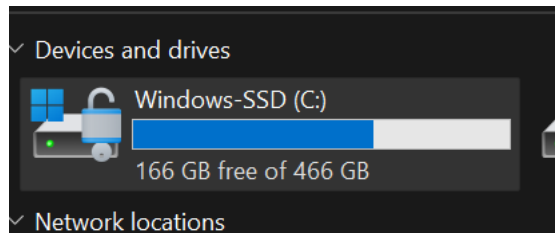
Task 3: Download VMware Workstation Pro and Ubuntu Server



Shows VMWare Workstation

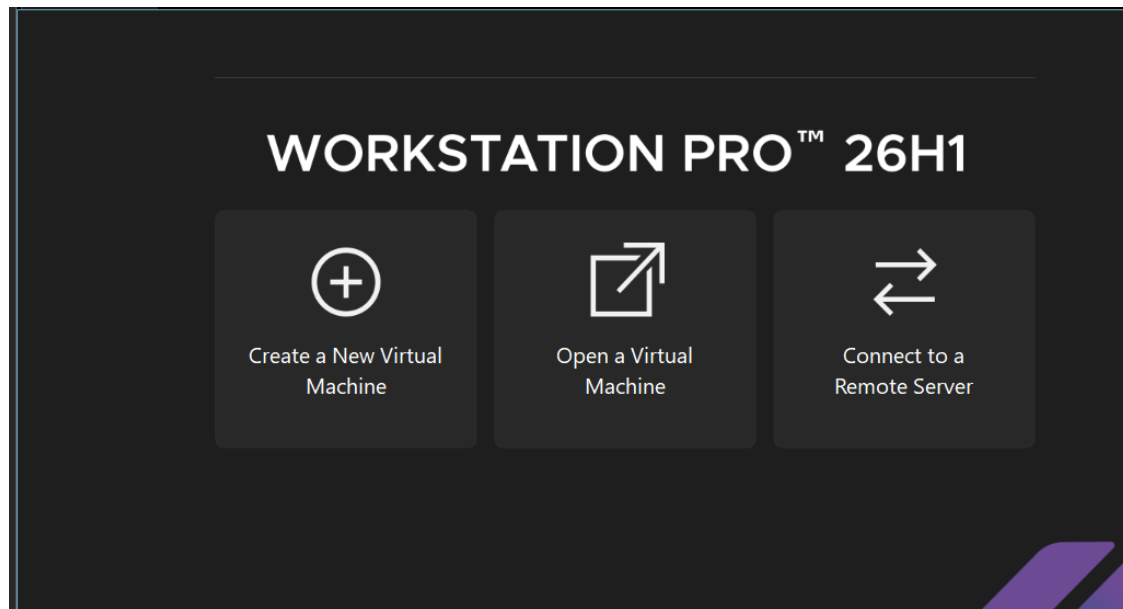
Name	Date modified	Type	Size
Today			
 ubuntu-26.04.1-live-server-amd64 (1)	9/12/2026 2:03 PM	ISO File	2,859,240 KB

Shows Ubuntu Server ISO File



Shows Available Storage

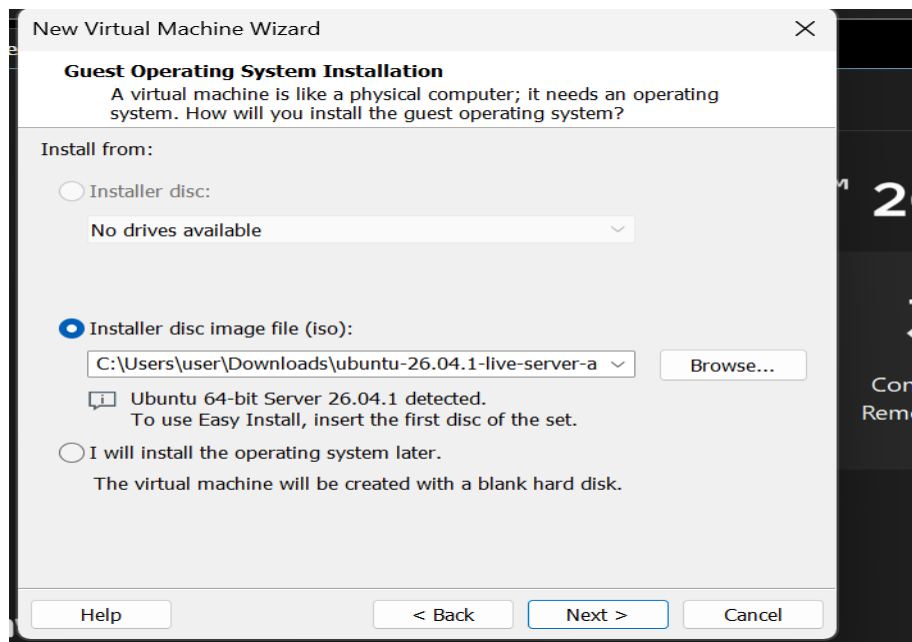
Task 4: Create the Ubuntu Server Virtual Machine



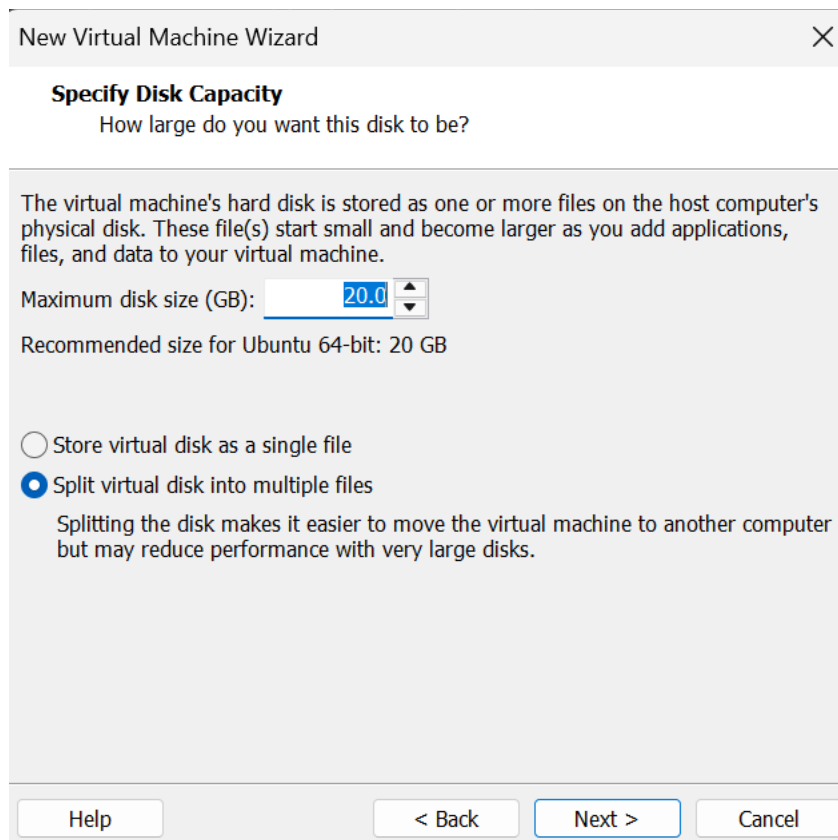
Shows the Creation of New Virtual Machine



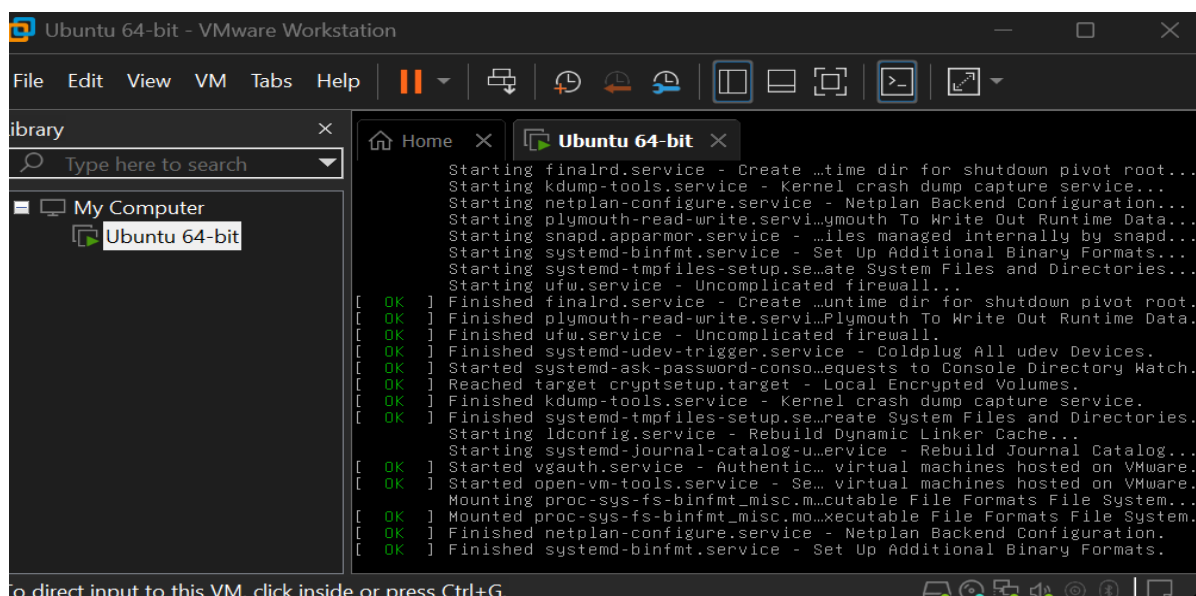
Shows the Selection of Typical (Recommended)



Shows the Selection of Ubuntu Server ISO

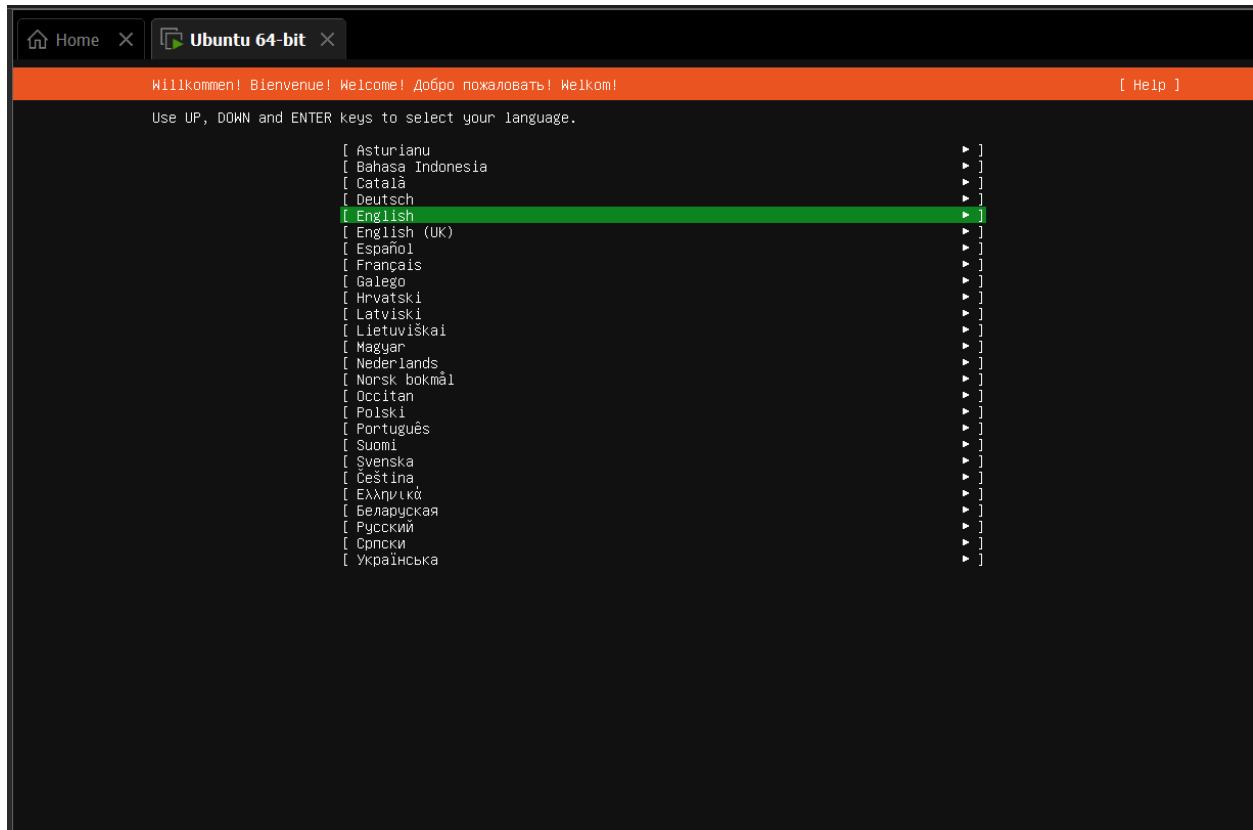


Shows VM Configuration Summary



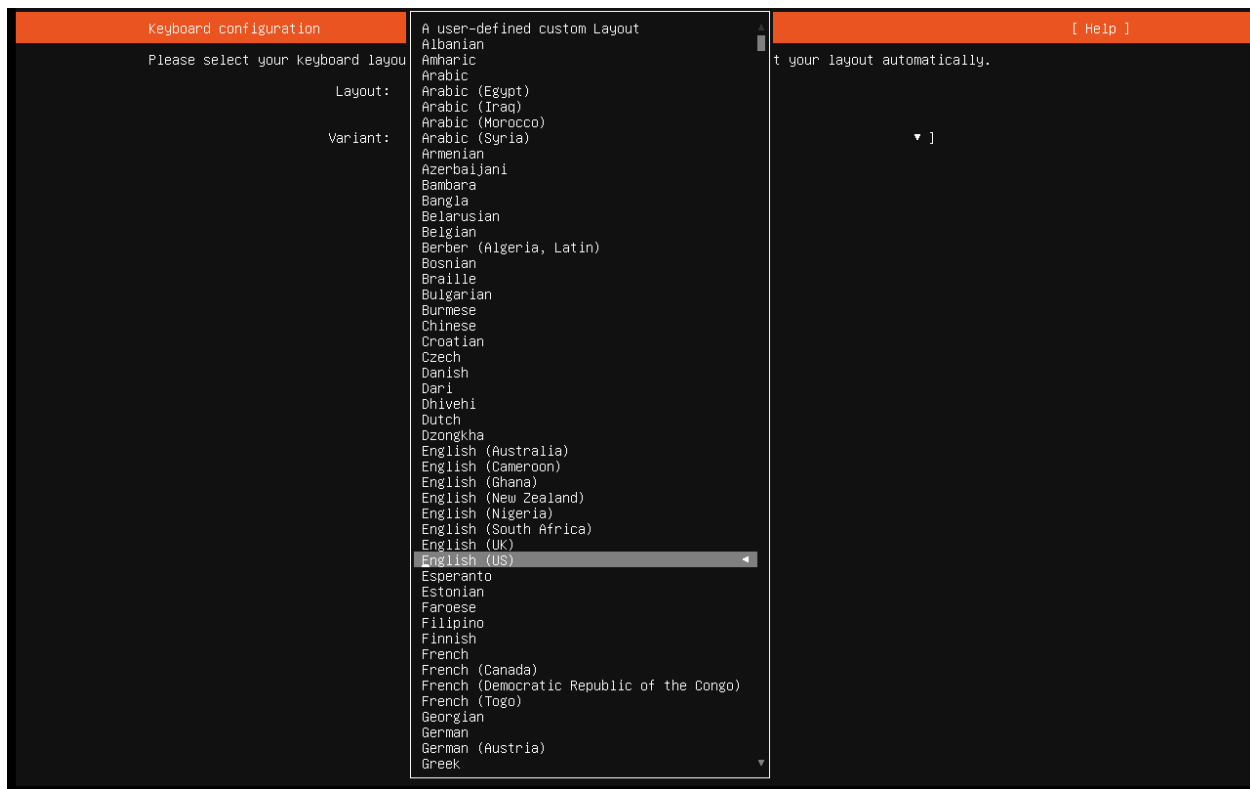
Shows Ubuntu Installer Boot

Task 5: Choose the Installation Language

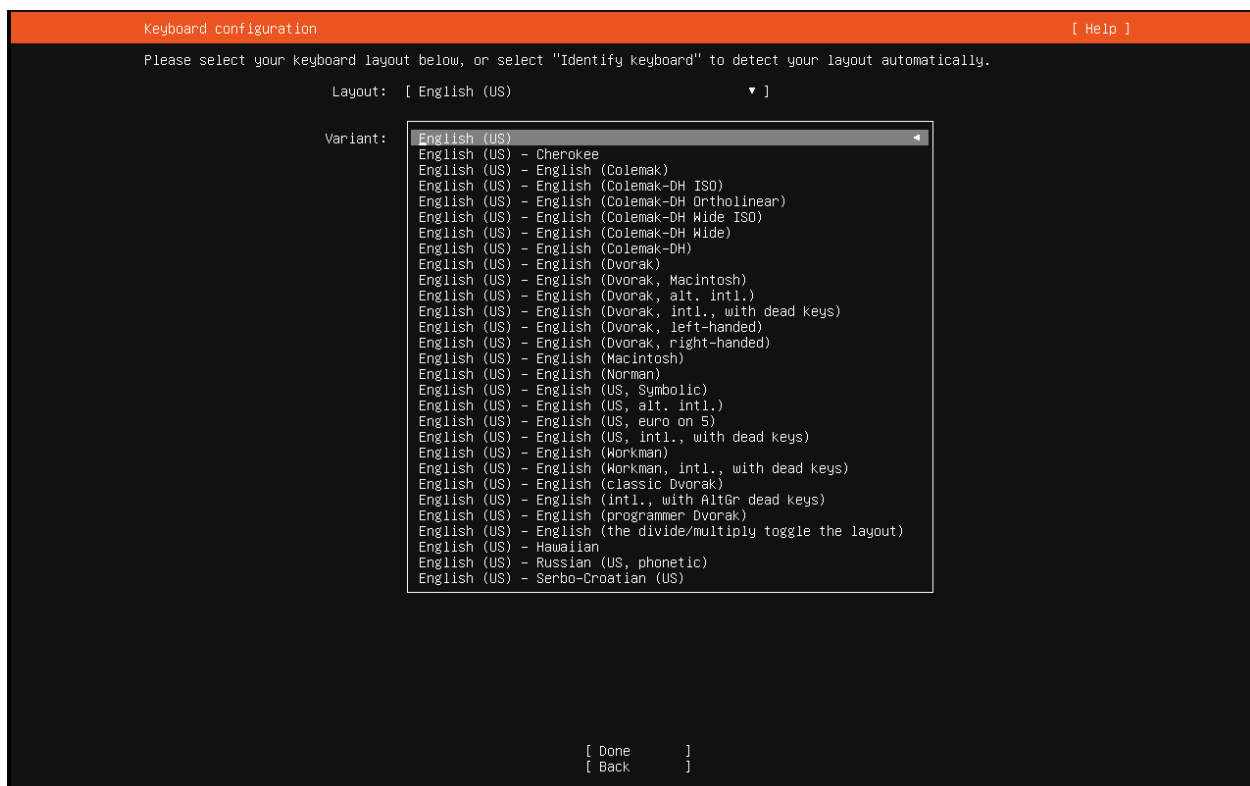


Shows Ubuntu Language Selection

Task 6: Configure the Keyboard Layout



Shows Keyboard Layout Selection



Shows Keyboard Variant Selection

Task 7: Choose the Ubuntu Server Installation Type

Choose the type of installation

[Help]

Choose the base for the installation.

☒ Ubuntu Server

The default install contains a curated set of packages that provide a comfortable experience for operating your server.

☐ Ubuntu Server (minimized)

This version has been customized to have a small runtime footprint in environments where humans are not expected to log in.

Additional options

☐ Search for third-party drivers

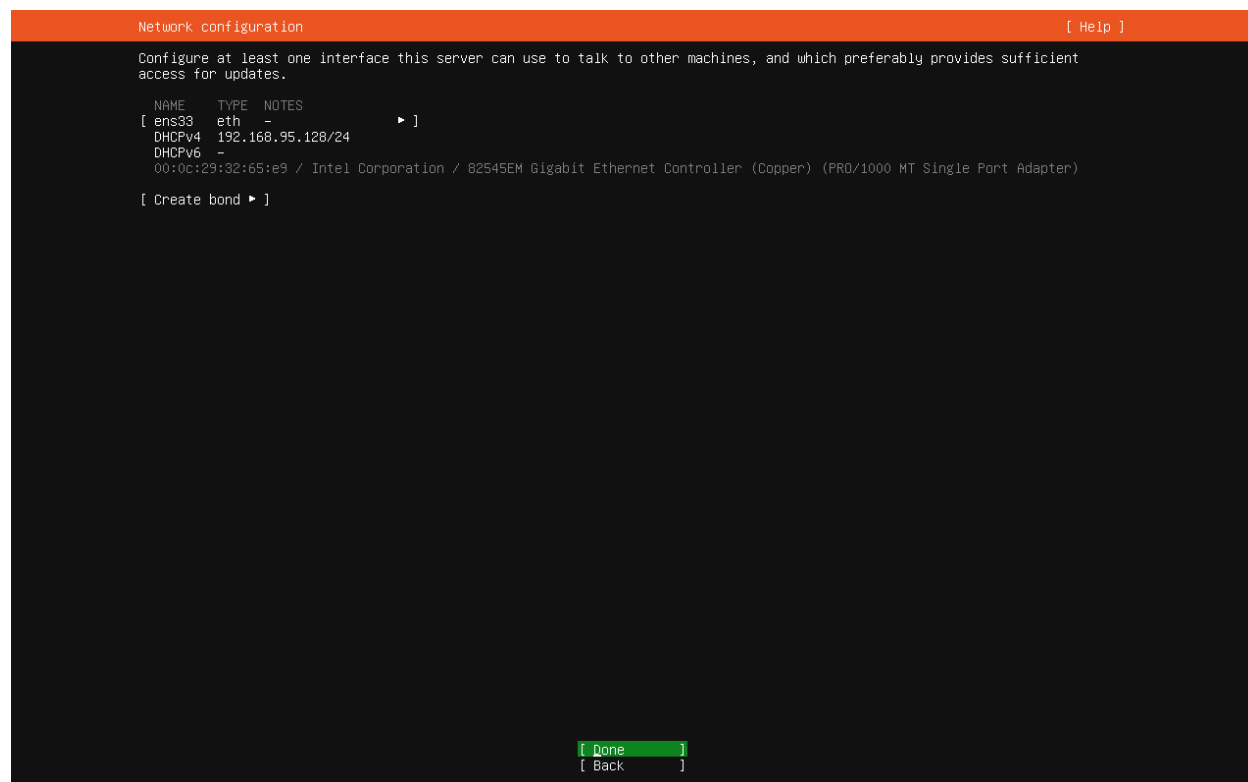
This software is subject to license terms included with its documentation. Some is proprietary. Third-party drivers should not be installed on systems that will be used for FIPS or the real-time kernel.

[Done]

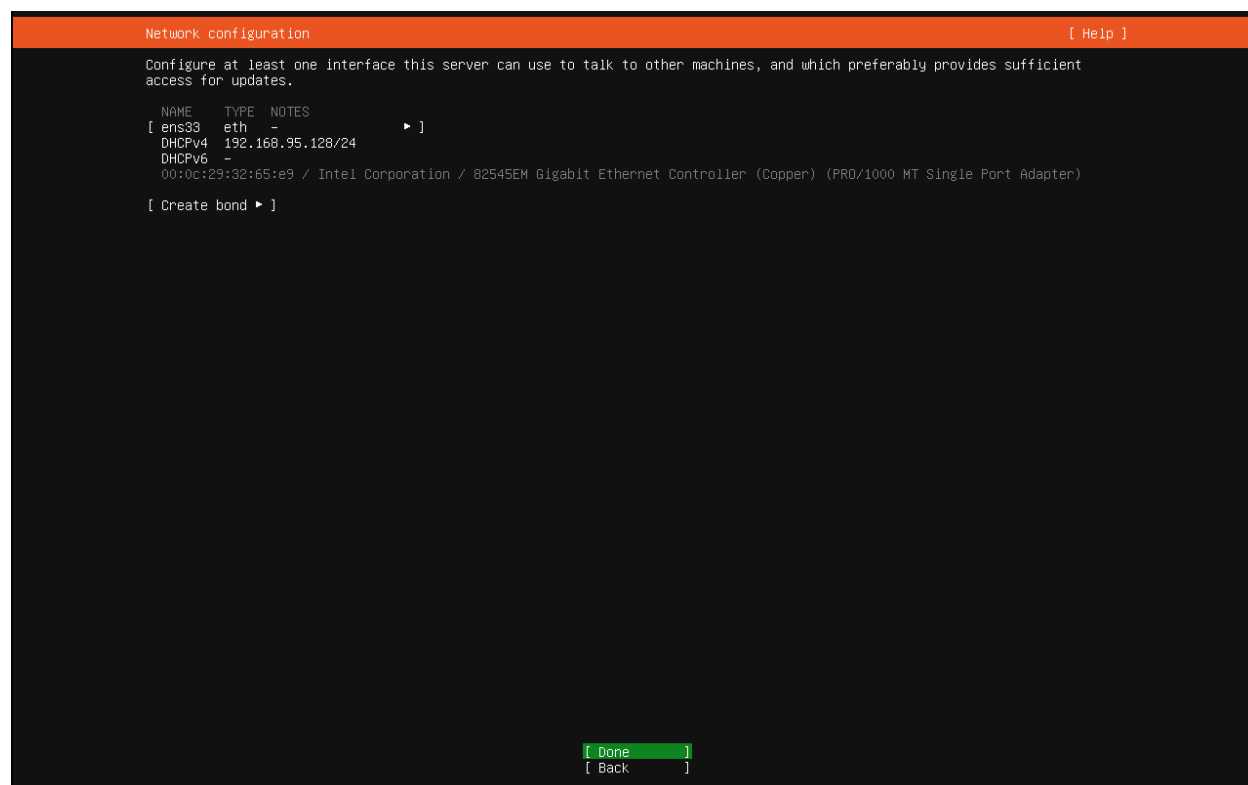
[Back]

Shows Ubuntu installation Type

Task 8: Configure the Network



Shows Ubuntu Network Interface



Shows Ubuntu Installer Network Interface

Task 9: Configure Storage

Guided storage configuration [Help]

Configure a guided storage layout, or create a custom one:

☒ Use an entire disk

[/dev/sda local disk 20.00GB ▼]

☒ Set up this disk as an LVM group

☐ Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

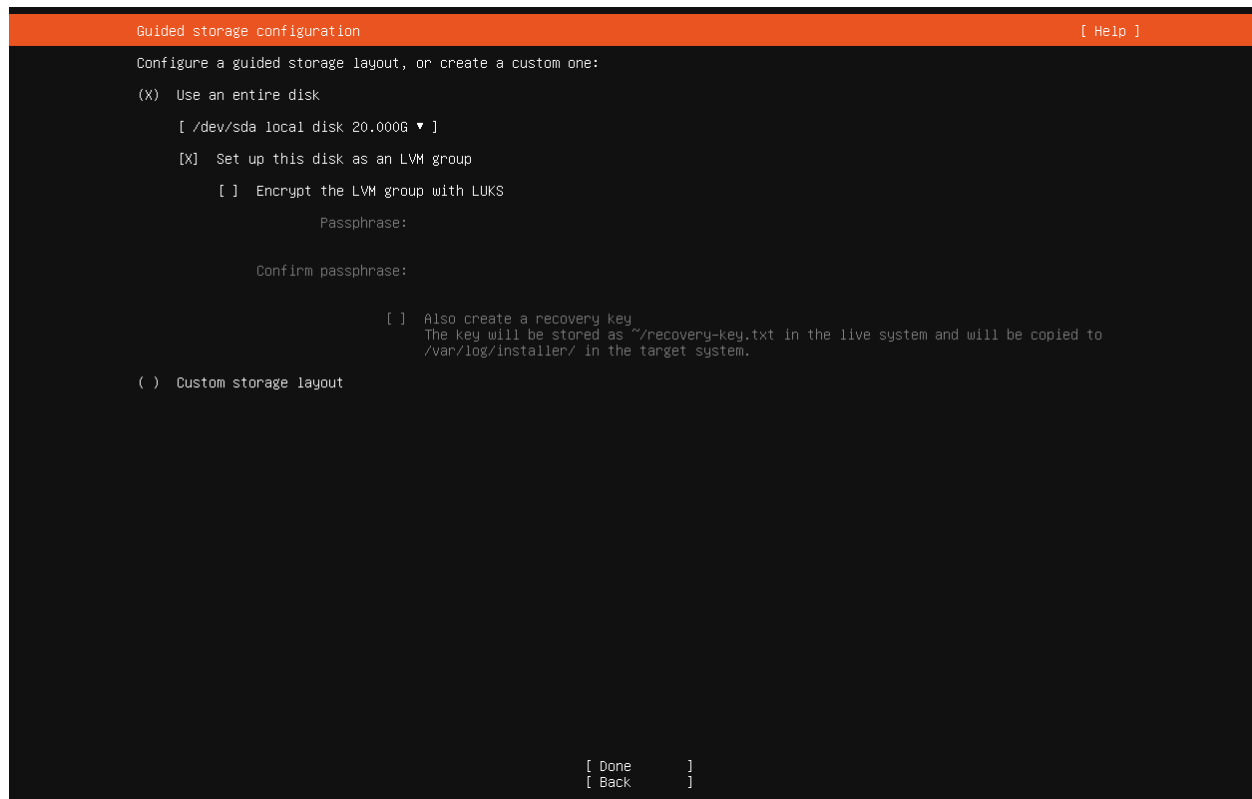
☐ Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

☐ Custom storage layout

[Done]

[Back]

Shows Ubuntu's Guided Storage



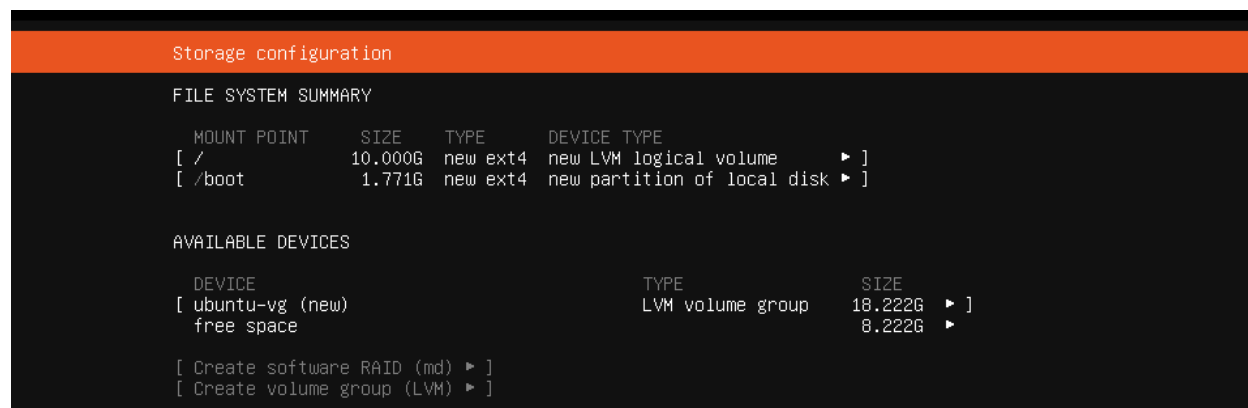
Shows Ubuntu's Storage Configuration

Task 10: Select the Installation Disk

```
Configure a guided storage layout, or create a custom one
(X) Use an entire disk
[ /dev/sda local disk 20.000G ▼ ]
```

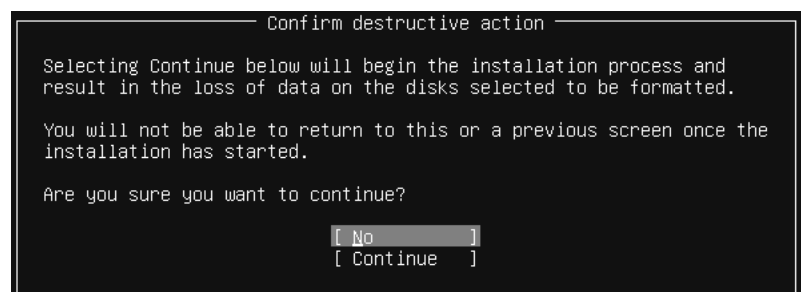
Shows Disk Selection Details

Task 11: Review the Partitions



Shows Ubuntu's Partition Layout

Task 12: Confirm the Storage Changes

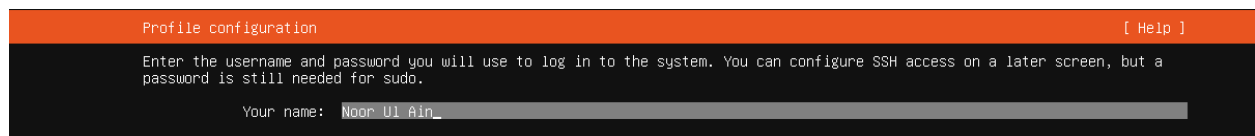


Shows Ubuntu's Storage Warning



Shows Ubuntu's Storage Confirmed

Task 13: Configure the Ubuntu Profile



Shows Ubuntu_Actual Name



Shows Ubuntu_Server Name



Shows Ubuntu_Username

Task 14: Complete the Software Installation

```

configuring format: format-0
configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpvt_ubsic/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
    configuring target system bootloader
    installing grub to target devices
    copying metadata from /cdrom
final system configuration
curtin command in-target
  calculating extra packages to install
  configuring cloud-init
  downloading and installing security updates
  curtin command in-target

```

Shows Ubuntu's Installation Progress

Task 15: Reboot and Verify the Ubuntu Identity

```

Installation complete! [ Help ]

  configuring mount: mount-0
  executing curtin install extract step
  curtin command install
    writing install sources to disk
    running 'curtin extract'
    curtin command extract
      acquiring and extracting image from cp:///tmp/tmpvt_ubsic/mount
  configuring keyboard
  curtin command in-target
  executing curtin install curthooks step
  curtin command install
    configuring installed system
    running 'curtin curthooks'
    curtin command curthooks
      configuring apt configuring apt
      installing missing packages
      configuring iscsi service
      configuring raid (mdadm) service
      configuring NVMe over TCP
      installing kernel
      setting up swap
      apply networking config
      writing etc/fstab
      configuring multipath
      updating packages on target system
      configuring pollinate user-agent on target
      configuring kernel crash dumps settings
      final kernel configuration
      configuring target system bootloader
      installing grub to target devices
      copying metadata from /cdrom
  final system configuration
  curtin command in-target
    calculating extra packages to install
    configuring cloud-init
    downloading and installing security updates
    curtin command in-target
      restoring apt configuration
      curtin command in-target
        subiquity/Late/run_builtins:
        subiquity/Late/run_user_supplied:

```

[View full log]
[Reboot Now]

Shows Ubuntu's Installation Completed

```

[ 20.636343] cloud-init[1146]: SHA256:RFYhIK0P+dYh24yUNlgYMNS9JyH9+yQoAXRk24DzfDo root@ubuntu
[ 20.636379] cloud-init[1146]: The key's randomart image is:
[ 20.636410] cloud-init[1146]: +---[RSA 3072]-----+
[ 20.636440] cloud-init[1146]: |  ++B*o+o.o. |
[ 20.636475] cloud-init[1146]: | ..++++.. |
[ 20.636505] cloud-init[1146]: | .o.oo. |
[ 20.636534] cloud-init[1146]: | =.X.* |
[ 20.636562] cloud-init[1146]: | *.%S* |
[ 20.636592] cloud-init[1146]: | E+.o |
[ 20.636622] cloud-init[1146]: | .... ... |
[ 20.636651] cloud-init[1146]: | .o |
[ 20.636683] cloud-init[1146]: | . |
[ 20.636713] cloud-init[1146]: +----[SHA256]-----+
[ 20.636741] cloud-init[1146]: Generating public/private ecdsa key pair.
[ 20.636773] cloud-init[1146]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 20.636804] cloud-init[1146]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 20.636835] cloud-init[1146]: The key fingerprint is:
[ 20.636861] cloud-init[1146]: SHA256:Yy15bdP7he300UtqiGtIWxQUcDMdx0doLUKJid1xhIA root@ubuntu
[ 20.636901] cloud-init[1146]: The key's randomart image is:
[ 20.636930] cloud-init[1146]: +---[ECDSA 256]-----+
[ 20.636959] cloud-init[1146]: | .#B@=+o. |
[ 20.636989] cloud-init[1146]: | E.=oB.+ |
[ 20.637053] cloud-init[1146]: | . * |
[ 20.637083] cloud-init[1146]: | + + o |
[ 20.637436] cloud-init[1146]: | . S + + . |
[ 20.637643] cloud-init[1146]: | + B . . + |
[ 20.637834] cloud-init[1146]: | = . . oo= |
[ 20.640608] cloud-init[1146]: | . . . .o+= |
[ 20.641062] cloud-init[1146]: | . . . .+ |
[ 20.643662] cloud-init[1146]: +----[SHA256]-----+
[ 20.645052] cloud-init[1146]: Generating public/private ed25519 key pair.
[ 20.645084] cloud-init[1146]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 20.646496] cloud-init[1146]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 20.647314] cloud-init[1146]: The key fingerprint is:
[ 20.648177] cloud-init[1146]: SHA256:C2CHSKZ2JW/os7QdBn7mrM41o1YIvx278A/D9b/c5uk root@ubuntu
[ 20.649741] cloud-init[1146]: The key's randomart image is:
[ 20.651456] cloud-init[1146]: +---[ED25519 256]-----+
[ 20.652724] cloud-init[1146]: | .+.o |
[ 20.653703] cloud-init[1146]: | o.*oo. |
[ 20.654491] cloud-init[1146]: | o = +.* |
[ 20.655303] cloud-init[1146]: | =. B o . |
[ 20.656014] cloud-init[1146]: | oo..+ S |
[ 20.656648] cloud-init[1146]: | ooo. . |
[ 20.657268] cloud-init[1146]: | o+Bo . |
[ 20.657973] cloud-init[1146]: | oo=o= o ... |
[ 20.658000] cloud-init[1146]: | .+ oo. +=E |
[ 20.662251] cloud-init[1146]: +----[SHA256]-----+

ubuntu login:
ubuntu login: _

```

Shows Ubuntu's Login Screen

```

[ 20.655303] cloud-init[1146]: | =. B o . |
[ 20.656014] cloud-init[1146]: | oo..+ S |
[ 20.656648] cloud-init[1146]: | ooo. . |
[ 20.657268] cloud-init[1146]: | o+Bo . |
[ 20.657973] cloud-init[1146]: | oo=o= o ... |
[ 20.658000] cloud-init[1146]: | .+ oo. +=E |
[ 20.662251] cloud-init[1146]: +----[SHA256]-----+

ubuntu login:
ubuntu login: noorulain027
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Sep 12 08:58:28 AM UTC 2026

System load:  0.04          Processes:           249
Usage of /:   50.5% of 9.75GB Users logged in:        0
Memory usage: 22%          IPv4 address for ens33: 192.168.95.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

noorulain027@ubuntu:~$ _

```

Shows Ubuntu's Terminal Login Screen

```
individual files in /usr/share/doc/*/*Copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
noorulain027@ubuntu:~$
```

Shows Ubuntu's Identity Verification

Task 16: Find the Ubuntu Server IP Address

```
applicable law.  
  
noorulain027@ubuntu:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000  
    link/ether 00:0c:29:5a:8f:3a brd ff:ff:ff:ff:ff:ff  
    altname enp2s1  
    altname enx000c295a8f3a  
    inet 192.168.95.130/24 metric 100 brd 192.168.95.255 scope global dynamic ens33  
        valid_lft 1489sec preferred_lft 1489sec  
    inet6 fe80::20c:29ff:fe5a:8f3a/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever  
noorulain027@ubuntu:~$
```

Shows Ubuntu Ip Command

```
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000  
    link/ether 00:0c:29:5a:8f:3a brd ff:ff:ff:ff:ff:ff  
    altname enp2s1  
    altname enx000c295a8f3a  
    inet 192.168.95.130/24 metric 100 brd 192.168.95.255 scope global dynamic ens33  
        valid_lft 1489sec preferred_lft 1489sec  
    inet6 fe80::20c:29ff:fe5a:8f3a/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever  
noorulain027@ubuntu:~$
```

Shows Ubuntu's IP Address

Task 17: Connect to Ubuntu Server from Windows

```
Microsoft Windows [Version 10.0.26200.9445]  
[c] Microsoft Corporation. All rights reserved.  
  
C:\Users\user>ssh noorulain027@192.168.95.130  
The authenticity of host '192.168.95.130 (192.168.95.130)' can't be established.  
ED25519 key fingerprint is SHA256:CZCHSKZzJW/osfQdBn7mrM4lo1YIvx278A/D9b/c5uk.  
This key is not known by any other names.
```

Shows Windows SSH Command

```
The authenticity of host '192.168.95.130 (192.168.95.130)' can't be established.  
ED25519 key fingerprint is SHA256:CZCHSKZzJW/osfQdBn7mrM4lo1YIvx278A/D9b/c5uk.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? YES  
Warning: Permanently added '192.168.95.130' (ED25519) to the list of known hosts.
```

Shows Windows SSH Fingerprint

```
The authenticity of host '192.168.95.130 (192.168.95.130)' can't be established.  
ED25519 key fingerprint is SHA256:CZCHSKZzJW/osfQdBn7mrM4lo1YIvx278A/D9b/c5uk.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? YES  
Warning: Permanently added '192.168.95.130' (ED25519) to the list of known hosts.  
noorulain027@192.168.95.130's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Sat Sep 12 09:30:51 AM UTC 2026  
  
System load:  0.01               Processes:            226  
Usage of /:   51.0% of 9.75GB    Users logged in:     0  
Memory usage: 24%               IPv4 address for ens33: 192.168.95.130  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
Updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

Shows Windows SSH Login

```
Expanded Security Maintenance for Applications is not enabled.
```

```
5 updates can be applied immediately.
```

```
To see these additional updates run: apt list --upgradable
```

```
Enable ESM Apps to receive additional future security updates.
```

```
See https://ubuntu.com/esm or run: sudo pro status
```

```
noorulain027@ubuntu:~$
```

Shows Windows SSH Identity